Uninsured people wait for basic health care at a free clinic in Los Angeles.

Can disparities be deadly?

Controversial research explores whether living in an unequal society can make people sick

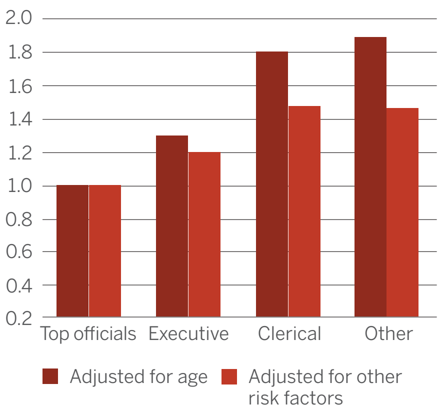
By **Emily Underwood**

Whitehall street, just south of Trafalgar Square in central London, is the heartbeat of the British government. Generations of workers in the highly stratified British Civil Service have marched to work each day in the government offices lining the road, with top bureaucrats working and living in palatial brick mansions built for aristocrats. Over the years, the denizens of Whitehall have fallen prey to the ills of the modern world: Their arteries have filled with fatty plaque; their blood sugar has spiked from diabetes; their lungs have been damaged by emphysema. And with surprising and troubling frequency, lower ranked workers have died earlier from these ailments than have their superiors.

To find out why, thousands of these civil servants, from typists to top officials, have gone to nearby medical clinics to have blood drawn, fill out questionnaires about how much they exercise and smoke, and don scratchy paper gowns for physical exams. Last year marked the 11th wave of data from this ambitious study, which has run for roughly 40 years and sparked an entire research program on the contentious question of whether being low-ranked can make you sick.

Deaths by rank at Whitehall

Relative rate of death over 25 years



Source: Marmot, 2000

HEALTHY AT THE TOP. In the long-running Whitehall studies, civil servants at every occupational grade live longer than their inferiors.

All agree that compared with the wealthy, poor people are less healthy. A child born in Norway can expect to live roughly 30 years longer than one born in Afghanistan. In the United States, on average, people in the highest income group can expect to outlive those in the lowest income group by more than 6 years. Preventable illnesses caused by poor nutrition and lack of education and care account for much of the disparity. Investing in health care and making it widely available can boost the health of those at the bottom. Redistributing wealth to the lower end of the curve helps, too. One simulation by researchers at the University of Otago, Wellington, for example, showed that shifting New Zealanders' incomes toward the mean income by 10% would save about 1100 lives per year.

But epidemiologist Michael Marmot of University College London (UCL), who leads the Whitehall study, argues that there's more to health than money alone. On the basis of his own and other studies, Marmot argues that hierarchy itself is a threat to health, with low-ranking individuals getting sicker and dying younger than higher-ups in part because of the sheer stress of being low on the social ladder.

Some public health experts say their own studies bear out Marmot's claim, but others think that confounding factors could easily account for the Whitehall findings. To these skeptics, focusing on hierarchy distracts from the real challenge of providing better health care to the poor. One

PHOTO: JOSEPH SOHM/VISIONS OF AMERICA/CORBIS

Published by AAAS

question is how being low on the social ladder matters to your health. Another is whether a society's health is worse when the rungs are far apart. The issue bubbles just below the surface in policy debates and has erupted recently in impassioned editorials. Some argue, paraphrasing Roman philosopher Seneca the Younger, that "to be poor in a wealthy society is the worst kind of poverty." But will it send you to an early grave?

WHO DIES FIRST? The Whitehall studies began as a simple search for heart disease risk factors. In the late 1960s, heart disease was thought to prey disproportionately upon upper-class, white-collar workers, because of their high-stakes jobs and type A personalities. After following more than 17,000 40- to 64-year-old male Whitehall employees for a decade, however, researchers at UCL found the opposite. During that period, 1652 men died, and men of the lowest rank were nearly four times more likely to die prematurely of heart disease than those in the highest tier, even though all had free health care.

In 1985, Marmot and his colleagues set out to determine why this might be so. They recruited a second cohort of more than 10,000 white-collar civil servants, including women, and found the same patterns of illness and mortality by rank, with some variations between men and women. Marmot started asking participants to fill out ever more extensive questionnaires, including not only their past medical history and health behaviors, but also their job demands, levels of stress, and social networks and support. As the data rolled in, he found that the psychological effects associated with status and job rank consistently predicted employees' health better than did their salaries, or even health-related behaviors like diet and exercise.

PODCAST

To hear a podcast with author Emily Underwood, see http://scim.ag/pod_6186.

Based on these findings, Marmot developed a theory: When a population moves beyond abject poverty, rank in the social hierarchy, not income, ultimately determines how healthy people are. Some animal studies suggest how status stress might "get under the skin," as epidemiologists put it: Low-ranking baboons and macaques can develop higher levels of stress hormones, atherosclerosis, and hypertension when subject to a dominant male's whims.

If Marmot and others are correct, simply shifting money to the poor won't be enough to boost their health. The health gradient among people who are not poor shows that it's "not only about poverty—we've got to improve society," he says.

From the dangerous streets of Chicago's South Side to the neatly tended homes of a Helsinki suburb, the link between low status and poor health has now been found in many different countries and contexts, says Ichiro Kawachi, a social epidemiologist at Harvard University. "The higher up the gradient you are, the longer you tend to live and the healthier you tend to be," he says.

SCALING UP. More controversial is whether overall population health is worse in more unequal societies. In 2009, Kawachi published a meta-analysis of epidemiological studies linking inequality and health in about 60 million people around the world. He and his colleague found an excess mortality risk of 8% for every 0.05 unit increase in a country's Gini coefficient, the most commonly used statistical measure of the gap between rich and the poor (see p. 818). Although such an effect may seem modest, when extrapolated to the global population it suggests that leveling income inequality could help avert more than 1.5 million deaths per year worldwide—assuming the effect is causal, he says.

In the United States, Kawachi and public health researcher S. V. Subramanian, also at Harvard, have found that income inequality is also strongly correlated with rates of infant mortality, heart disease, and several health conditions across many states and cities, even after controlling for variables such as absolute income in each location, race, age, and education. Measures of social cohesion such as trust also appear to track with inequality, he says. In one of America's most unequal states, Louisiana, for example, people are far more likely to agree with the statement that "most people would try to take advantage of you if they got the chance."

Based on such studies, Kawachi and others argue that inequality breaks down social values, such as trust and support, that protect against both physical and mental illness. In a recent op-ed in *The New York Times*, epidemiologists Richard Wilkinson and Kate Pickett of the University of York in the United Kingdom took the argument even further. They claim that the reason more unequal countries like the United States see higher rates of schizophrenia and other mental illnesses is because inequality causes "social corrosion" that damages the individual psyche.

Others aren't convinced. John Lynch, an epidemiologist at the University of Adelaide in Australia, says that although he started out as a "true believer" in the income inequality hypothesis, a string of negative and equivocal studies turned him into a skeptic. Back in a 2004 paper, for example, Lynch and colleagues reviewed 98 cross-national studies and found "little evidence" of a consistent link between income inequality and health, although the United States displayed a more robust association than others. Working on well-established public health goals such as reducing smoking and improving the living conditions of the poor will likely have more direct health impacts than targeting relative income gaps, he says.

Even if the correlations Kawachi and others have found hold up, there's no strong evidence that income inequality, per se, is directly damaging people's health, says Angus Deaton, an economist at Princeton University. In American cities and states where there are large proportions of African-Americans, for example, racism, poor health care, and political disenfranchisement could just as easily explain poor health outcomes as income inequality, he says. Deaton argues that extreme in-

The gap between what people desire—like these luxury cars in South Africa—and what they can afford may be a source of unhealthy stress.

PHOTO: SCHALK VAN ZUYDAM/AP/CORBIS

equality is a risk to health chiefly because it skews politics to favor the rich and powerful in society. "I get angry" over Wilkinson's claim that psychological stress is the primary culprit, because it completely deflects from the real issues," he says.

CAUSE OR CORRELATE? In 2011, Princeton University economists Christina Paxson and Anne Case found another potential explanation for the correlations between rank and health. They reexamined data from the Whitehall II study and found that adults who were healthier as children started at higher grades in the Civil Service, were promoted to higher positions, and maintained better health throughout their lives. Occupational rank was a marker, but not a cause, of poor health in adulthood, Paxson and Case concluded.

Many economists agree that people's health influences their status, rather than the other way around, says Dalton Conley, a sociologist at New York University in New York City. "Economists tend to think that your health predicts where you are on the social scale," he says. If you're sick a lot and miss school, for example, you won't do as well in the labor market. He notes that the initial Whitehall studies also didn't take into account "very controversial" questions about the extent to which genes determine later health and wealth.

Marmot says he's now persuaded that genetics and early-life experiences do play some role in adult health and socioeconomic rank. Still, neither can fully account for the huge difference in mortality and morbidity among Whitehall's occupational grades, he says. Pointing to more than 100 studies based on Whitehall data, Marmot maintains that stressors such as lack of control and harassment at work fall hardest on low-ranking workers and take a fatal toll.

Causality lies at the heart of the issue, so scientists are now looking for mechanisms that could link inequality and health. Biocultural anthropologist Elizabeth Sweet of the University of Massachusetts, Boston, notes that any causal link assumes that people know their place in the hierarchy. "We don't always walk around with our salaries tacked on our foreheads, so how do we get the information to make that social comparison?"

Kawachi suggests that for Americans, their own aspirations may provide the point of comparison. Even though an American born in the bottom fifth of the income distribution has only about an 8% chance of rising to the top fifth—half the likelihood of a child born in Denmark—more than 90% of Americans still believe in the American dream, he says, and the collision of their ideal with reality may take a toll on health. "When you work hard on the assumption that we're building a meritocracy, then fail," the resulting depression and frustration may contribute to the country's high rates of drug abuse, suicide, and violence, he says.

Similarly, Sweet hypothesizes that the gap between the standard of consumption one identifies with success and one's ability to meet that ideal produces measurable stress and health impacts. Through extensive interviews, she and others collect information about the cultural norms of material success in a given community. In rural Brazil, being successful might mean owning a TV, whereas in U.S. suburbs it might mean having the "right" brand of jeans or cellphone. The researchers measure the degree to which an individual is able to "keep up with the Joneses," and compare that with health indicators such as the amount of cortisol in saliva, a marker of stress.

In a study of African-American teenagers in Chicago, Sweet demonstrated that teens who could easily conform to their communities' "ideal" level of consumption had lower blood pressure than teens who couldn't meet those norms. But if they managed to get expensive sneakers and brand-name clothes even though they couldn't really afford them, the students had abnormally high blood pressure. In Sweet's view, this suggests that the tension caused by the gap between what people need and what they can afford can affect health. But she and Kawachi admit that the causal link is tenuous.

Back at Whitehall, civil servants are still striding into work every morning. Some of the original participants have retired and moved to the suburbs. Many others have died, leaving behind reams of data about what they ate, if they exercised, and how often they felt lonely. Marmot and others have produced more than 500 papers based on these workers' experiences and continue to churn out dozens each year. To fully explain the links between inequality, rank, and health, however, may take hundreds more. ■